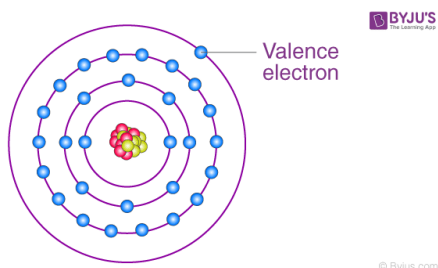


2nd semester

EEE-175: Electronics

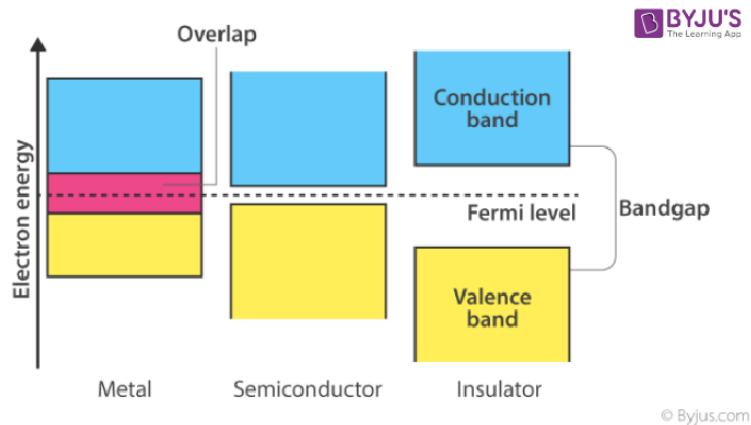
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- **Valence electrons:** Valence electrons are the s and p electrons in the outermost shell.



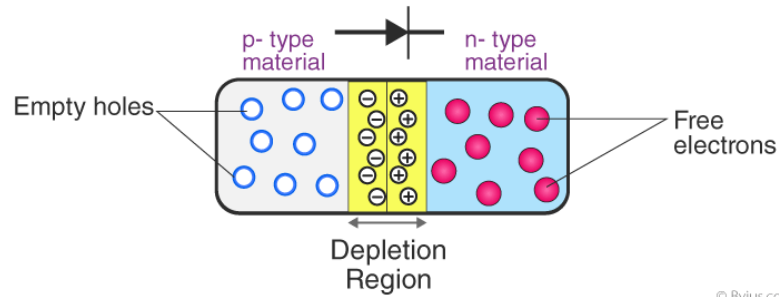
- **Valence Band and Conduction Band:** Here are some of the differences between a valence band and a conduction band.

Conduction Band	Valence Band
1. Higher energy level band	1. An energy band is formed by a series of energy levels containing valence electrons
2. Partially filled by the electrons	2. Always filled with electrons
3. High energy state	3. Low energy state
4. Above the Fermi level	4. Below of Fermi level
5. Electrons move into the conduction band when the atom is excited	5. Electrons will move out of the conduction band when the atom is excited
6. Current flows due to such electrons	6. The highest energy level, which can be occupied by an electron in the valence band at 0 K



- **Semiconductor material:** A semiconductor material is a type of material that has electrical conductivity that is intermediate between a conductor (such as metal) and an insulator (such as glass). Semiconductors can be pure elements, such as silicon or germanium. Semiconductor materials are essential components in electronic devices such as transistors, diodes, and integrated circuits. They are used in a wide range of applications, including computers, smartphones, solar cells, and many other electronic devices.
- **p-n material:** A p-n material refers to a semiconductor material that has been doped with both p-type and n-type impurities to create a p-n junction. The p-n junction is a boundary or interface between the p-type and n-type regions inside a semiconductor crystal. The p-type region has an excess of holes (positive charge carriers), while the n-type region has an excess of electrons (negative charge carriers). This imbalance of charge carriers at the p-n junction creates unique electrical properties.
- **p-n junction:** A P-N junction is an interface or a boundary between two semiconductor material types, namely the p-type and the n-type, inside a semiconductor.

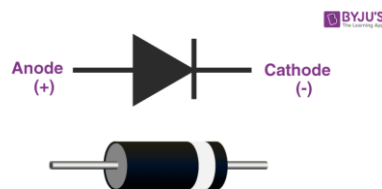
UNBIASED P-N JUNCTION



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- **Diode:** A diode is a two-terminal electronic component that conducts electricity primarily in one direction. It has high resistance on one end and low resistance on the other end.

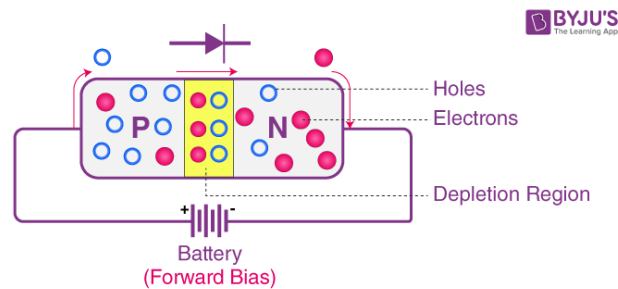


- **Semiconductor Diode:** A semiconductor diode is a **p-n junction diode**. It is a two-terminal device that conducts current only in one direction.

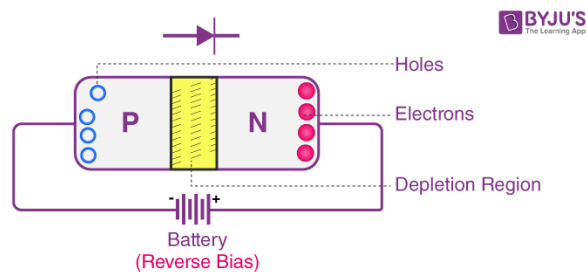


- **P-N Junction Diode Under Forward Bias:** When we apply the external voltage across the semiconductor diode in such a way that the p-side is connected to the positive terminal of the battery and the n-side is connected to the negative terminal, then the semiconductor diode is said to be forward-biased. In this case, the built-in potential of the diode and thus the width of the depletion region decreases, and the height of the barrier gets reduced. The overall barrier voltage, in this case, comes out to be $V_0 - V$, which is the

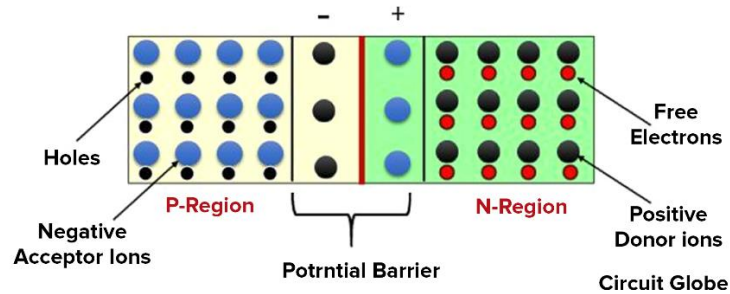
difference between the built-in potential and the applied potential. As we supply a small amount of voltage, the reduction in the barrier voltage from the above-given formula is very less and thus only a small number of current carriers cross the junction in this case.



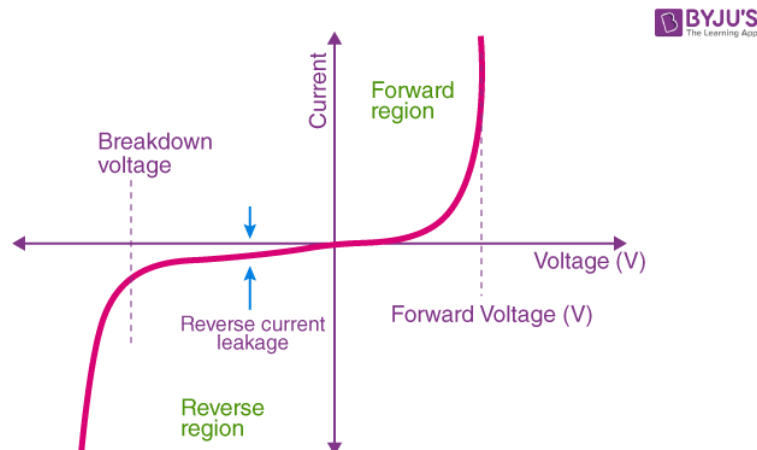
- **P-N Junction Diode Under Reverse Bias:** When we apply the external voltage across the semiconductor diode in such a way that the positive terminal of the battery is connected to its n-side and the negative terminal of the battery is connected to the p-side of the diode, then it is said to be in the condition of reverse bias. When an external voltage is applied across the diode, as the direction of the external voltage is the same as that of the barrier potential, the total voltage barrier sums up to be $(V_0 + V)$.



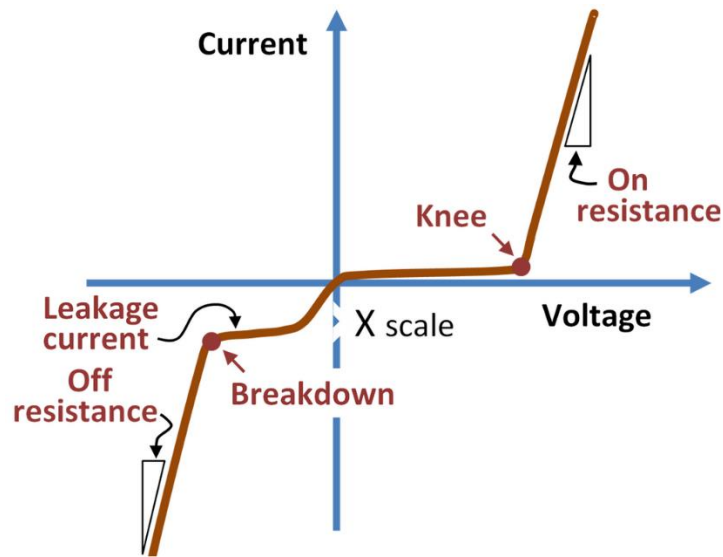
- **Potential Barrier:** The potential barrier in the PN-junction diode is the barrier in which the charge requires additional force for crossing the region. In other words, the barrier in which the charge carrier stopped by the obstructive force is known as the potential barrier.



- **Semiconductor Diode Characteristics:** The V-I characteristics curve of a semiconductor diode is given below. This characteristic curve is a typical explanation for the V-I characteristic of a semiconductor diode. Current in the semiconductor diode starts to conduct when the current exceeds the threshold of the forward voltage, which is mentioned by the manufacturer.



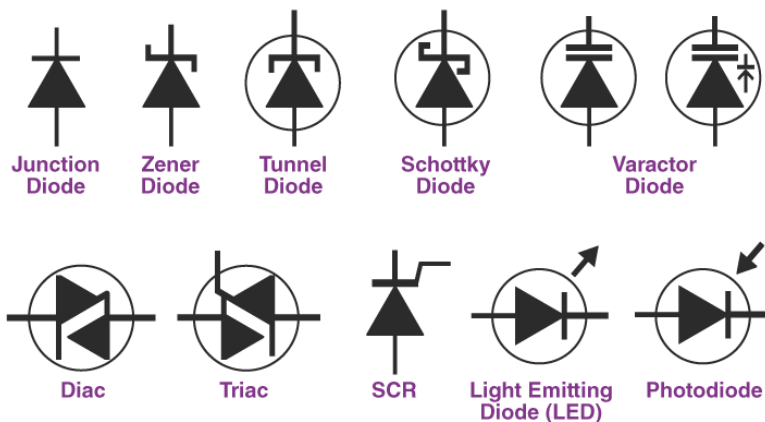
- **Knee voltage:** The "knee voltage" refers to the voltage at which a diode starts conducting current when it is forward biased. In a typical silicon diode, the knee voltage is around 0.7 volts, while in a germanium diode, it is around 0.3 volts. When the forward bias voltage across the diode exceeds the knee voltage, the diode begins to conduct current.



- **Leakage current:** The "leakage current" refers to the small amount of current that flows through a diode when it is reverse biased. Even though a diode is designed to block current in the reverse direction, there is still a small amount of current that can flow due to imperfections in the diode's structure. This leakage current is typically in the microampere (μA) range and can increase with higher temperatures.

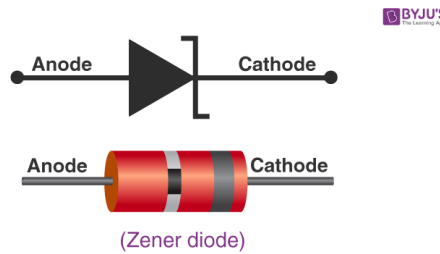
TYPES OF DIODES

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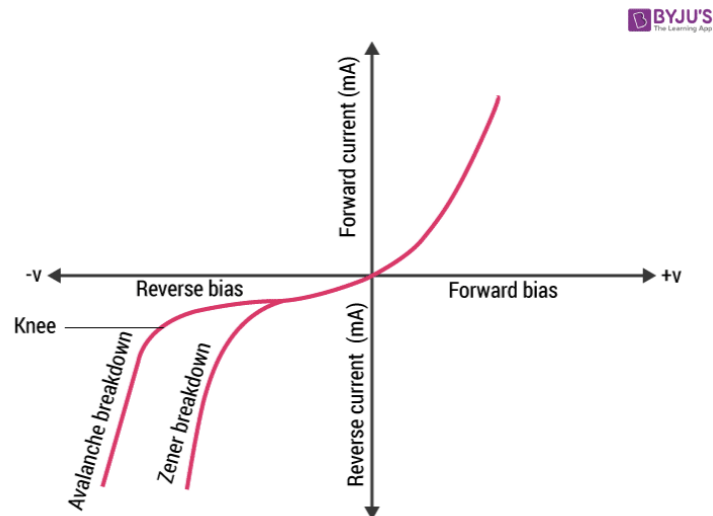
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- **Zener Diode:** A Zener diode is a highly doped semiconductor device specifically designed to function in the reverse direction. It is engineered with a wide range of Zener voltages (V_z), and certain types are even adjustable to achieve variable voltage regulation.



- **V-I Characteristics of Zener Diode:**

The diagram given below shows the V-I characteristics of the Zener diode.



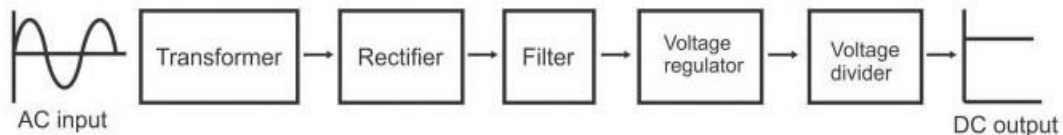
The V-I characteristics of a Zener diode can be divided into two parts as follows:

- Forward Characteristics
- Reverse Characteristics

- **Zener/Breakdown Voltage** – The Zener or the reverse breakdown voltage ranges from 2.4 V to 200 V, sometimes it can go up to 1 kV while the maximum for the surface-mounted device is 47 V.

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- **Unregulated Power Supply:** An unregulated power supply is one whose dc terminal voltage is affected significantly by the amount of load. As the load draws more current, the dc terminal voltage becomes less.
- **Regulated Power supply:** It is that dc power supply whose terminal voltage remains almost constant regardless of the amount of current drawn from it.



A typical DC power supply consists of 5 stages as shown in Fig:

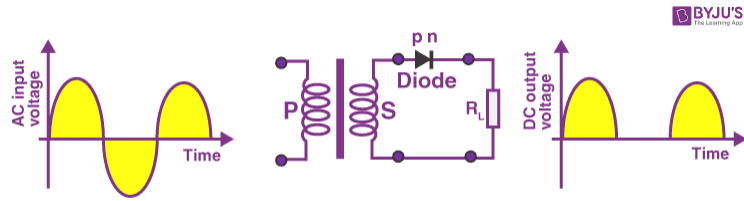
- **Transformer:** It is responsible for stepping up or stepping down the input voltage to the desired level. The transformer consists of primary and secondary windings and operates on the principle of electromagnetic induction. It isolates the input from the output and provides the necessary voltage level for further processing.
- **Rectifier:** It is a circuit which employs one or more diodes to convert AC voltage into **pulsating dc voltage**.
- **Filter:** After rectification, the output of the rectifier contains ripple or variations in the DC voltage. The filtering stage is used to smooth out the output voltage by removing the ripple. It typically involves the use of capacitors and inductors to filter out the AC components and provide a more stable DC voltage. The filtered output is closer to the desired regulated DC voltage.

- **Voltage Regulator:** The voltage regulator stage is responsible for maintaining a constant output voltage regardless of changes in the input voltage or load conditions. It uses voltage regulation circuits such as voltage regulators or feedback control loops to regulate the output voltage. The voltage regulator continuously monitors the output voltage and adjusts it as needed to compensate for any variations, ensuring a stable and regulated DC voltage.
 - **Voltage Divider:** The voltage divider stage is often included in regulated DC power supplies to further adjust the output voltage to the desired level. It consists of resistors connected in series to divide the voltage and provide precise control over the output voltage. By selecting the appropriate resistor values, the voltage divider allows for fine-tuning of the output voltage to meet specific requirements.
- **Pulsating DC (AC মিশ্রিত DC):** Pulsating DC refers to a type of direct current (DC) electrical signal that oscillates or varies in amplitude over time. Unlike traditional DC, where the current flows in a constant direction at a constant voltage, pulsating DC experiences changes in value but not in direction. Pulsating DC can be produced from alternating current (AC) using a half-wave rectifier or a full-wave rectifier. The rectifier converts the AC waveform into a pulsating DC waveform by allowing only one or both halves of the AC waveform to pass through, depending on the type of rectifier used. This results in a waveform that consists of one or more rectangular pulses rather than a continuous sinusoidal waveform.
- **Halfwave rectifier(2131 page):** A halfwave rectifier is defined as a type of rectifier that allows only one-half cycle of an AC voltage waveform to pass while blocking the other half cycle.

Working of Half Wave Rectifier:

Let's understand how a half-wave rectifier transforms AC into DC.

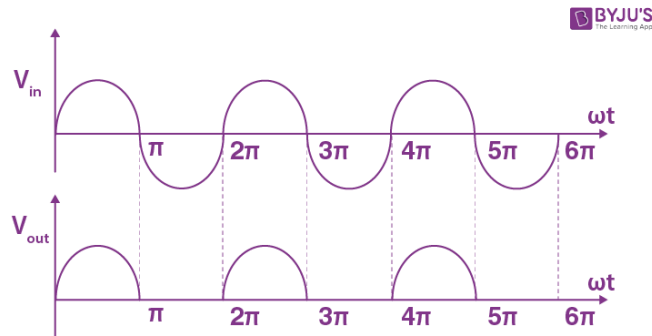
- A high AC voltage is applied to the primary side of the step-down transformer. The obtained secondary low voltage is applied to the diode.
- The diode is forward biased during the positive half cycle of the AC voltage and reverse biased during the negative half cycle.
- The final output voltage waveform is as shown in the figure below:



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Half Wave Rectifier Waveform:

The halfwave rectifier waveform before and after rectification is shown below in the figure:



Half Wave Rectifier Formula:

Ripple Factor:

Ripple factor determines how well a halfwave rectifier can convert AC voltage to DC voltage.

Ripple factor can be quantified using the following formula:

$$\gamma = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1}$$

The ripple factor of a halfwave rectifier is 1.21.

Efficiency:

The efficiency of a halfwave rectifier is the ratio of output DC power to the input AC power.

The efficiency formula for halfwave rectifier is given as follows:

$$\eta = \frac{P_{DC}}{P_{AC}}$$

RMS value:

The RMS value of the load current for a half-wave rectifier is given by the formula:

$$I_{rms} = \frac{I_m}{2}$$

Form factor:

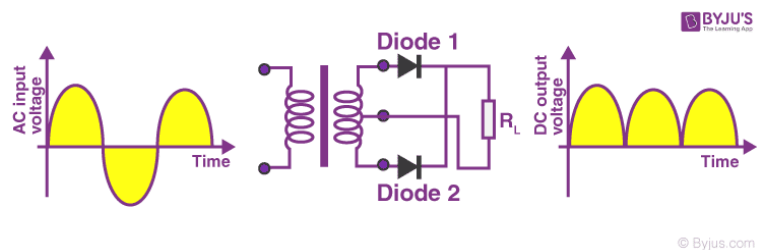
The form factor is the ratio between RMS value and average value and is given by the formula:

$$\text{Form Factor} = \frac{\text{RMS Value}}{\text{Average Value}}$$

- **Full-wave rectifier(2136 page):** A full wave rectifier is defined as a rectifier that converts the complete cycle of alternating current into pulsating DC. Unlike halfwave rectifiers that utilize only the halfwave of the input AC cycle, full wave rectifiers utilize the full cycle. Full-wave rectifier circuits are used for producing an output voltage or output current which is purely DC.

Working of Full Wave Rectifier:

The full-wave rectifier utilizes both halves of each AC input. When the p-n junction is forward biased, the diode offers low resistance and when it is reverse biased it gives high resistance. The circuit is designed in such a manner that in the first half cycle if the diode is forward biased then in the second half cycle it is reverse biased and so on.

**Full-Wave Rectifier Formula:****Ripple Factor:**

Ripple factor for a full-wave rectifier is given as:

Efficiency:

Rectifier efficiency is used as a parameter to determine the efficiency of the rectifier to convert AC into DC. It is the ratio of DC output power to the AC input power. The rectifier efficiency of a full-wave rectifier is **81.2%**.

$$\eta = \frac{DC \text{ Output Power}}{AC \text{ Output Power}}$$

RMS value:

The RMS value of the current can be calculated using the following formula:

$$I_{rms} = \frac{I_{max}}{\sqrt{2}}$$

Form factor:

The form factor of the full wave rectifier is calculated using the formula:

$$K_f = \frac{RMS \text{ value of current}}{Average \text{ value of current}} = \frac{I_{rms}}{I_{dc}} = \frac{I_{max}/\sqrt{2}}{2I_{max}/\pi} = \frac{\pi}{2\sqrt{2}} = 1.11$$

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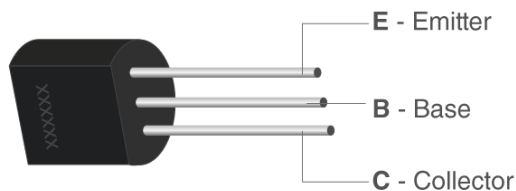
- **Transistor:** A transistor is a type of semiconductor device that can be used to conduct and insulate electric current or voltage. A transistor basically acts as a switch and an amplifier. In simple words, we can say that a transistor is a miniature device that is used to control or regulate the flow of electronic signals.

Parts of a Transistor:

A typical transistor is composed of three layers of semiconductor materials or, more specifically, terminals which help to make a connection to an external circuit and carry the current. A voltage or current that is applied to any one pair of the terminals of a transistor controls the current through the other pair of terminals. There are three terminals for a transistor. They are listed below:

- **Base:** This is used to activate the transistor.
- **Collector:** It is the positive lead of the transistor.
- **Emitter:** It is the negative lead of the transistor.

PARTS OF A TRANSISTOR



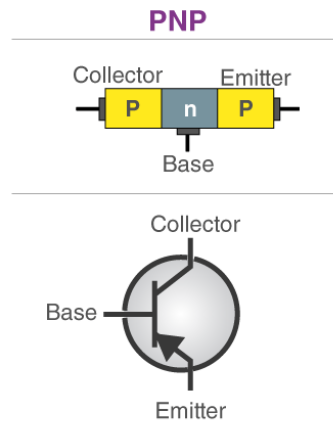
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- **Bipolar Junction Transistor(BJT):** A bipolar junction transistor is a three-terminal semiconductor device that consists of two p-n junctions which are able to amplify or magnify a signal. It is a current controlled device. The three terminals of the BJT are the base, the collector, and the emitter. A signal of a small amplitude applied to the base is available in the amplified form at the collector of the transistor. This is the amplification provided by the BJT.

There are two types of bipolar junction transistors:

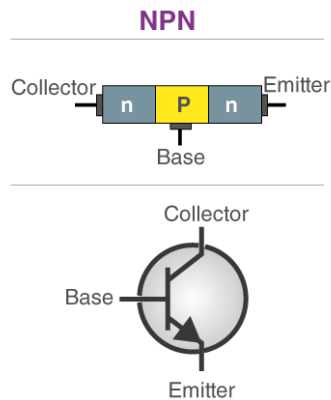
- **PNP bipolar junction transistor:** In PNP BJT, the n-type semiconductor is sandwiched between the two p-type semiconductors. The two p-type

semiconductors act as emitter and collector respectively while the n-type semiconductor acts as a base. This is shown in the figure below.



The current enters the transistor through the emitter such that the emitter-base junction is forward biased and the collector-base junction is reverse biased.

- **NPN bipolar junction transistor:** In NPN BJT, p-type semiconductor is sandwiched between the two n-type semiconductors. The two n-type semiconductors act as emitter and collector respectively while the p-type semiconductor acts as a base. This is shown in the figure below.



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Current entering the emitter, base, and collector has the sign convention of positive while the current that leaves the transistor has the sign convention of negative.